PORT DETECTION

When using HW2, detection pins must be connected using one of the following methods.

1. Current mirror circuit for dynamic/hot-plug port detection (Deprecated): Voltage drop is a ~0.3V in idle. This method is unstable during rumble motor activation, as the high current draw causes a voltage drop below USB specifications, leading to controller disconnection. While the circuit performs perfectly fine with rumble disabled in the configuration, it is currently not recommended for full-feature use. Although it offers better isolation and safety, it requires extra components, additional wiring, and more effort to implement.

2. Port GND isolation – dynamic / hot-plug port detection
Requires one 10 kΩ resistor per port.
Disconnect (pull out) the ground wire (black) from the mainboard connector for each gameport that will be used. Connect this wire together with the corresponding port detection pin and add a 10 kΩ pull-up resistor to the ESP 3.3 V rail.

3. Port shield isolation – dynamic / hot-plug port detection
Requires one 10 kΩ resistor per port.
Place isolation (e.g. Kapton or insulating tape) between the metal shield of each used port and the Xbox chassis shield. Solder an additional wire to the port shield and connect the other end together with the corresponding port detection pin and a 10 kΩ pull-up resistor to ESP 3.3 V.

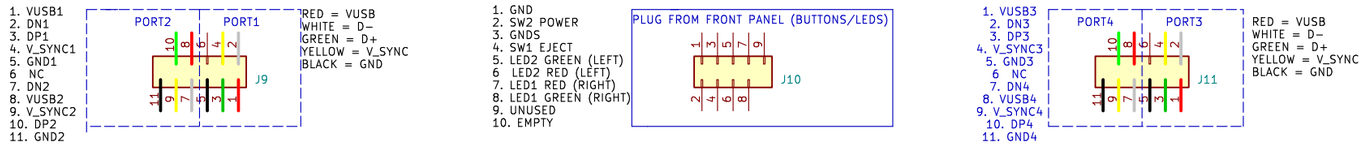
4. Pull-up only – no detection, static port count
Pull up the detection pin to the ESP 3.3 V power rail via a resistor.
To enable a port for BT devices, one 10 kΩ resistor per port is required.

Note: Wired gamepad support on all ports enabled in this way is permanently disabled.

Important
If detection pins are not pulled up using one of the methods above, the BT controller cannot be used on the corresponding port.
For all unused detection ports, to avoid false state activation – especially on GPIO32 and GPIO33, as they are highly sensitive touch pins – the following method must be used:

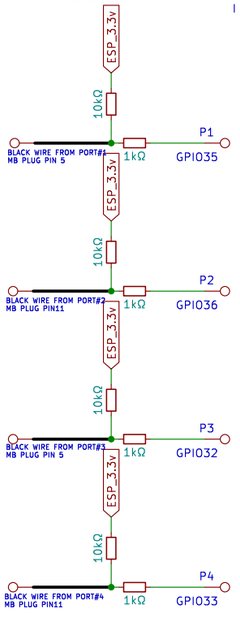
5. Pulldown – no detection, port disabled
Unused detection pins must be pulled down to GND through a 10 kΩ resistor.

GAMEPORTS PLUGS FROM FRONT PANEL – TOP VIEW AT MAINBOARD

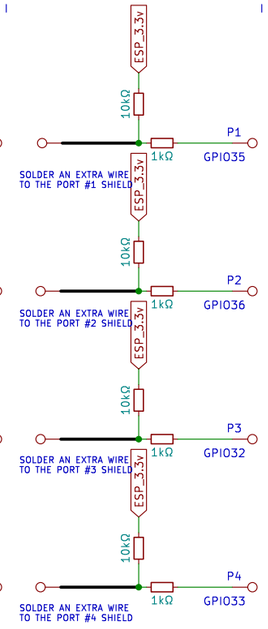


PLAYER1=PORT1/GPIO35
PLAYER2=PORT2/GPIO36
PLAYER3=PORT3/GPIO32
PLAYER4=PORT4/GPIO33

2.PORT GND ISOLATION DYNAMIC DETECTION

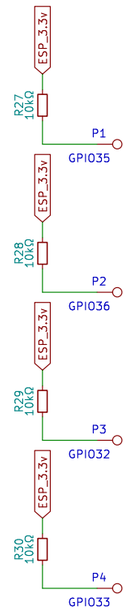


3.PORT SHIELD ISOLATION DYNAMIC DETECTION

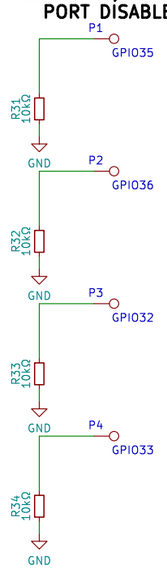


1 kΩ series resistors on GPIOs are not required – they provide only additional GPIO protection.

4. PULLUP ONLY / NO DETECTION STATIC PORT COUNT



5.PULLDOWN / NO DETECTION PORT DISABLED FOR BT

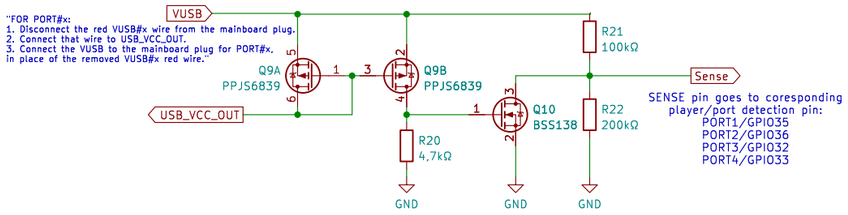


1.CURRENT MIRROR CIRCUIT DYNAMIC DETECTION

Design by ManCloud at https://github.com/ManCloud/CurrentTrigger
Instalation examples at https://github.com/nostalgic-indulgences/BlueRetro_Internal_Installation

One circuit for one port.

TODO: working but
probably lower pullups needed
investigate i2c bus



R1 – 4.7kΩ 0805 SMD Resistor
R2 – 100kΩ 0805 SMD Resistor
Q1 – YJS2301A SOT-23-6 DUAL P-Channel MOSFET
Q2 – BSS123 / BSS138 / A2SHB SOT-23 N-Channel MOSFET

This project allow using BT gamepads on ogXBOX
BlueRetro by DarthCloud https://github.com/darthcloud/BlueRetro
OGX360 by Ryzee119 https://github.com/Ryzee119/ogx360
Based on concept and code by netham45 https://github.com/netham45/BlueRetro
BROGX by Konwektor https://github.com/konwektor/BlueRetro

Sheet: /HW2_PORT_DETECTION/
File: PORT_DETECTION_sch.kicad_sch

Title: BROGX		
Size: A3	Date: 2026-03-01	**Rev: 1.6**
KiCad E.D.A. 9.0.2		Id: 5/11